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Circuit-noise-resilient virtual distillation

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Xiao-Yue Xu, Chen Ding, Shuo Zhang, Wan-Su Bao & He-Liang Huang

Quantum error mitigation (QEM) is vital for improving quantum algorithms' accuracy on noisy near-term devices. A typical QEM method, called Virtual Distillation (VD), can suffer from imperfect implementation, potentially leading to worse outcomes than without mitigation. To address this, we introduce Circuit-Noise-Resilient Virtual Distillation (CNR-VD), which includes a calibration process using simple input states to enhance VD's performance despite circuit noise, aiming to recover the results of an ideally conducted VD circuit. Simulations show that CNR-VD significantly mitigates noise-induced errors in VD circuits, boosting accuracy by up to tenfold over standard VD. It provides positive error mitigation even under high noise, where standard VD fails. Furthermore, our estimator's versatility extends its utility beyond VD, enhancing outcomes in general Hadamard-Test circuits. The proposed CNR-VD significantly enhances the noise-resilience of VD, and thus is anticipated to elevate the performance of quantum algorithm implementations on near-term quantum devices.

In the pursuit of large-scale fault-tolerant quantum computers, the presence of noise arising from environmental noise and imperfect hardware remains a significant challenge^{1–4}. The field of Quantum error mitigation (QEM) techniques^{5–13} braves this challenge, offering hopeful solutions that reduce the impact of noise even in the absence of full-blown quantum error correction^{14–17}. Among these techniques, one typical QEM method called virtual distillation (VD)^{18–25} has gained considerable attention. For a given N -qubit noisy quantum state ρ and an observable O , the n -order VD algorithm aims to extract the ideal component from ρ using n copies of ρ . VD offers distinct advantages for exponential suppression of errors in expectation value estimation for observables, and its state-noise-agnostic nature.

However, employing the VD algorithm for high-accuracy expectation values encounters two primary challenges. The first challenge is the coherent mismatch^{18,19,26}, which can be mitigated through combinations of other QEM methods such as quantum subspace expansion^{11,27,28} in the state preparation circuit of ρ ¹⁰, or polynomial fitting of multiple orders of VD¹⁹. The second challenge arises from noise in the VD circuit. The Hadamard-Test-like VD circuit¹⁹ utilized to efficiently implement VD consists of controlled derangement operator D_n and controlled operator O . As the number of multi-qubit gates in the VD circuit increases linearly with nN ¹⁹, this can introduce non-negligible deviations in the estimated expectation values, potentially surpassing the initial errors in state preparation. Mitigating this error often involves treating the VD circuit as a part of the target circuit and leveraging additional QEM techniques¹⁹, or employing shadow estimations^{29–31} to avoid executing the VD circuit, albeit with an exponential measurement overhead^{20,24}.

In this work, we focus on mitigating the noise in the VD circuit and propose the circuit-noise-resilient virtual distillation (CNR-VD). In our method, we integrate a calibration procedure that utilizes the noisy VD

circuit. During this procedure, the noisy state ρ undergoes a transformation into easily-prepared states that are independent of ρ . The expectation values of these states are obtained via noisy VD (VD with a noisy VD circuit), and are subsequently employed to enhance mitigation outcomes, which is facilitated by a function derived from noisy VD. Numerical results confirm the effectiveness of CNR-VD, demonstrating a significant improvement in accuracy—up to an order of magnitude—compared to noisy VD. We also observe that CNR-VD raises the gate noise threshold for VD, resulting in effective mitigation outcomes. Moreover, we compare the accuracy and efficiency of CNR-VD with two common VD-based methods designed to counter noise in the VD circuit, and observe the superiority of CNR-VD.

Results

Virtual distillation with noisy VD circuit

Suppose we have an N -qubit noisy quantum state ρ which can be expressed as

$$\rho = (1 - \epsilon)|\psi\rangle\langle\psi| + \epsilon \sum_k \lambda_k |\psi_k\rangle\langle\psi_k| \quad (1)$$

through spectral decomposition where $\epsilon \geq 0$ is the error rate, along with the ideal state $|\psi\rangle\langle\psi|$ and the noisy components $\sum_k \lambda_k |\psi_k\rangle\langle\psi_k|$ (satisfying $\sum_k \lambda_k = 1$) of ρ . The n -order VD algorithm produces mitigated expectation value of observable O as

$$\langle O \rangle_{\text{VD}} = \frac{\text{Tr}[\rho^n O]}{\text{Tr}[\rho^n]} \approx \langle \psi | O | \psi \rangle + \mathcal{O}(\epsilon^n), \quad (2)$$

where the error term $\mathcal{O}(\epsilon^n)$ is exponentially suppressed with n . By means of the VD circuit, we do not require the physical preparation of ρ^n . The VD

circuit consists of the controlled derangement operator D_n containing nN CSWAP gates and the controlled operator O . Accordingly, the mitigated value is

$$\langle O \rangle_{\text{VD}} = \frac{\text{Tr}[\rho^n O]}{\text{Tr}[\rho^n]} = \frac{\text{Tr}[D_n((\rho O) \otimes \rho^{\otimes n-1})]}{\text{Tr}[D_n \rho^{\otimes n}]}, \quad (3)$$

where $\text{Tr}[\rho^n O]$ is calculated by $2p_0(\rho, O) - 1$ with the probability $p_0(\rho, O)$ of measuring the ancilla qubit Q_0 in the zero state. We can use the same method to estimate $\text{Tr}[\rho^n]$ by replacing O with I in the VD circuit.

Now we will analyze the effects of the noise in the VD circuit. We consider the k -weight ($k \leq N$) Pauli observable $O = \prod_{\omega=1}^k O_{i_\omega}^{\omega}$ where the ω -th non-trivial Pauli term on qubit Q_{i_ω} is $O_{i_\omega}^{\omega} \in \{\sigma_x, \sigma_y, \sigma_z\}$. Then the controlled operator O is comprised of k two-qubit controlled Pauli gates. For the gate noise assumption, we primarily focus on the m -qubit ($m \geq 1$) stochastic Pauli gate noise $\mathcal{E}_m^p$ with error rate $0 < p_m < 1$,

$$\mathcal{E}_m^p(\rho) = (1 - p_m)\rho + p_m \sum_{s=1}^{4^m-1} \delta_s^m P_s^m \rho P_s^m, \quad (4)$$

where P_s^m is the s -th Pauli operator on m qubits and the coefficients $\{\delta_s^m\}_s$ satisfy $\delta_s^m \geq 0$, $\sum_s \delta_s^m = 1$. Considering that measurement error mitigation protocols^{32–38} such as Tensor Product Noise (TPN)^{35,39} can effectively suppress single-qubit readout errors in VD, which purely involves measurements on one qubit, we set up clean readout of the ancilla. We denote $\tilde{p}_0(\rho, O)$ to be the noisy version of $p_0(\rho, O)$, which is contaminated by the noise in the VD circuit, and we will pay our attention to the second-order VD. Then the outcome gained through noisy VD circuit can be expressed as a function

$$2\tilde{p}_0(\rho, O) - 1 = \sum_{j=0}^{\lfloor \frac{k}{2} \rfloor} \sum_{t=0}^{\binom{k}{j}} -1 a_{j,t}^k(p_1, p_2, p_3, O) A_j(t), \quad (5)$$

where j denotes the number of unwanted SWAP gates generated by σ_x/σ_y Pauli errors occurring on the controlled qubit of CSWAP gates, and t represents the t -th combination of positions for these j SWAPs. And $a_{j,t}^k(p_1, p_2, p_3, O)$ refers to the ρ -independent parameters for $A_j(t)$ (we define $A_0(0) = A_k(0) = \text{Tr}[\rho^2 O]$), which can be written as

$$A_j(t) = (1 - \epsilon)^2 \text{Tr}[O_1^{j,t} |\psi\rangle\langle\psi|] \cdot \text{Tr}[O_2^{j,t} |\psi\rangle\langle\psi|] + \epsilon^2 \sum_k \lambda_k^2 \text{Tr}[O_1^{j,t} |\psi_k\rangle\langle\psi_k|] \cdot \text{Tr}[O_2^{j,t} |\psi_k\rangle\langle\psi_k|]. \quad (6)$$

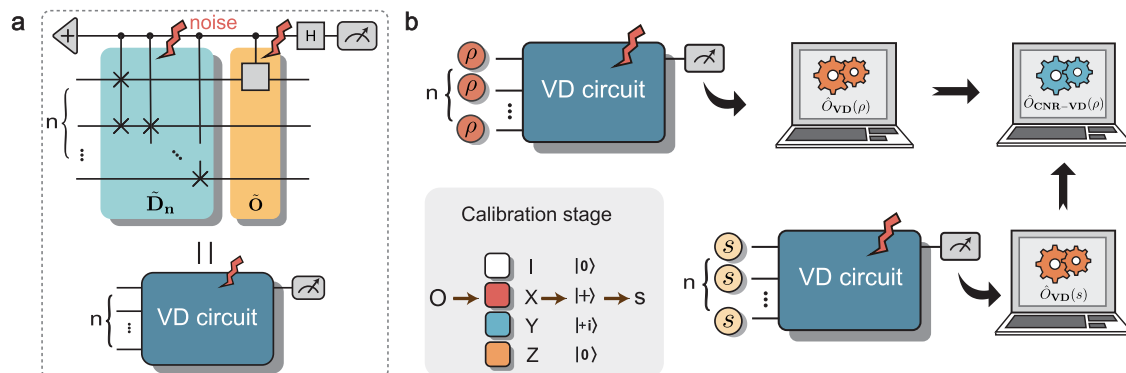


Fig. 1 | The noisy virtual distillation (VD) circuit and the circuit-noise-resilient virtual distillation (CNR-VD). **a** The noisy VD circuit of n -order VD to estimate the expectation value of Pauli observable O . **b** The illustration of CNR-VD. In the n -order noisy VD (VD with noisy VD circuit), we use n copies of the noisy state ρ to compute the estimator $\hat{O}_{\text{VD}}(\rho)$. The CNR-VD estimator $\hat{O}_{\text{CNR-VD}}(\rho)$ integrates a

$A_j(t)$ is a combination of expectation values of observables $O_1^{j,t}$ and $O_2^{j,t}$ which meets $O = O_1^{j,t} + O_2^{j,t}$, and it derives from the t -th combination of j undesired SWAPs (See more details in the Supplementary Note 2).

Circuit-noise-resilient virtual distillation

We introduce the CNR-VD, which aims to extract $\text{Tr}[\rho^2 O]$ attained by ideal VD (VD with noiseless VD circuit) based on the function in Eq. (5). We begin with the simplest case involving a one-weight Pauli observable O (equally applicable to I), which holds $A_j(t) = \text{Tr}[\rho^2 O]$ for $j = 0, 1$ and $t = 0$. Then Eq. (5) reduces to

$$2\tilde{p}_0(\rho, O) - 1 = a(p_1, p_2, p_3, O) \text{Tr}[\rho^2 O]. \quad (7)$$

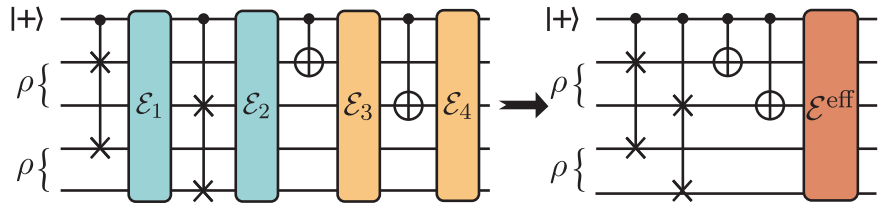
The ρ -independent parameter $a(p_1, p_2, p_3, O)$ is amenable to calibration if we assume the temporal stability of Pauli error rate p_i ($i = 1, 2, 3$). In order to obtain these parameters, we add a calibration procedure using the noisy VD circuit where the noisy state ρ is replaced by some separate state s , which can be easily prepared via single-qubit gates, as shown in the Fig. 1b. The state s can be any eigenstate of O with eigenvalue $+1$ which satisfies $\text{Tr}[s^2 O] = 1$, and the parameter $a(p_1, p_2, p_3, O)$ in Eq. (7) can be estimated by $2\tilde{p}_0(s, O) - 1$ with an estimation error $\mathcal{O}(p_1)$ due to the noisy preparation of s . We can also attain the estimation of parameter $a(p_1, p_2, p_3, I)$ for $\text{Tr}[\rho^2]$ through a noisy VD circuit with $O = I$. Then with the noisy VD estimator defined as $\hat{O}_{\text{VD}}(\rho) = (2\tilde{p}_0(\rho, O) - 1)/(2\tilde{p}_0(\rho, I) - 1)$, we can establish the CNR-VD estimator

$$\hat{O}_{\text{CNR-VD}}(\rho) = \frac{\left(\frac{2\tilde{p}_0(\rho, O) - 1}{2\tilde{p}_0(s, O) - 1}\right)}{\left(\frac{2\tilde{p}_0(\rho, I) - 1}{2\tilde{p}_0(s, I) - 1}\right)} = \frac{\hat{O}_{\text{VD}}(\rho)}{\hat{O}_{\text{VD}}(s)}, \quad (8)$$

which exhibits resilience to the noise in the VD circuit for any one-weight Pauli observable O . We can also employ the CNR-VD estimator to obtain reliable expectation values for higher-weight Pauli observables under moderate noise level of Pauli gate noise, with an estimation error $\mathcal{O}(p_1, p_2, p_3)$ as we ignore terms in Eq. (5) with $j > 1$. When the assumption of stochastic Pauli gate noise model in our work does not suit the practical scenario, the employment of randomized compiling^{40–42} (RC) technique can enable us come close to the stochastic Pauli gate noise. Moreover, since the independence on ρ of the parameters in Eq. (7), estimations of multiple noisy states can be carried out once the calibration procedure is executed to minimize circuit resource consumption, as long as the temporal stability of gate noise holds at least during a certain time period.

calibration procedure where the noisy state ρ undergoes a transformation into an easily-prepared state s . The expectation value $\hat{O}_{\text{VD}}(s)$ is obtained via noisy VD and subsequently employed to enhance mitigation outcomes. In the calibration procedure highlighted in the gray box in **b**, we show one possible choice for s which is the eigenstate of Pauli O with eigenvalue $+1$.

Fig. 2 | The effective noise of the second-order VD circuit for the two-qubit noisy state ρ and two-weight Pauli observable $\sigma_x^{\otimes 2}$. We define the general global Markovian noise of the i -th layer as $\mathcal{E}_{i+1}$. After relocating all the noise to the end of the VD circuit, we define the mixture of these noise conjugated by the layers behind as the effective noise $\mathcal{E}^{\text{eff}}$.



Besides the local Pauli gate noise model, our CNR-VD estimator is also applicable to the general Markovian noise model without restrictions on locality. For the n -order VD, suppose the VD circuit has l layers of noisy gates $\mathcal{U}_{1:l} = \prod_{j=1}^l \mathcal{E}_j^j \mathcal{U}_j$, where $\mathcal{E}_j^j$ is the Markovian noise acting on the j -th layer. After relocating all the noise to the end of the VD circuit, we define the mixture of these noise conjugated by the layers behind as the effective noise $\mathcal{E}^{\text{eff}}(\cdot) = \sum_i K_i \cdot K_i^\dagger$ of the VD circuit, where K_i is the Kraus operator of the effective noise. Hence, the noisy VD circuit can be divided into an ideal VD circuit followed by the effective noise, as shown in Fig. 2. We define an $(nN + 1)$ -qubit state $\rho^{\text{RD}} = \rho^{\text{R}} \otimes \rho^{\text{D}}$ where ρ^{R} is a single-qubit pure state and ρ^{D} is an (nN) -qubit operator. Then if the effective noise $\mathcal{E}^{\text{eff}}$ satisfies any of these two conditions:

- There exists a set of Kraus operators such that: each Kraus operator $K_i = K_i^{\text{R}} \otimes K_i^{\text{D}}$ of $\mathcal{E}^{\text{eff}}$ is separate into subsystems ρ^{R} and ρ^{D} , along with unitary K_i^{R} and K_i^{D} .
- $\mathcal{E}^{\text{eff}} = \mathcal{E}^{\text{eff,R}} \otimes \mathcal{E}^{\text{eff,D}}$, along with the single-qubit unital noise channel $\mathcal{E}^{\text{eff,R}}$ and the CPTP noise channel $\mathcal{E}^{\text{eff,D}}$.

The CNR-VD estimator $\hat{O}_{\text{CNR-VD}}(\rho)$ can reduce the error in the expectation value induced by $\mathcal{E}^{\text{eff}}$ when the state preparation error for s is negligible compared to $\mathcal{E}^{\text{eff}}$ (See proofs in the Supplementary Note 3). Since the depolarizing gate noise satisfies Condition 2 as it commutes with the CSWAP gates in the VD circuit, $\hat{O}_{\text{CNR-VD}}(\rho)$ shows high tolerance to low-error-rate Pauli gate noise, which is close to the depolarizing gate noise. We also establish other forms of the CNR-VD estimator with loosened requirement of $\mathcal{E}^{\text{eff}}$ in the Supplementary Note 3.

Furthermore, we can extend our CNR-VD to encompass a broader range of quantum computing applications beyond QEM algorithms. By generalizing the operators D_n and O in the VD circuit to arbitrary Hermitian operator U , we can get a typical Hadamard-Test circuit. Then we can use the CNR-VD estimator to calculate the expectation value of $\text{Tr}[\rho U]$ and reduce the additional noise caused by the controlled- U . As for the input state s used in the calibration procedure, we can choose any state that satisfies $\text{Tr}[sU] = 1$ (See more details and simulations in Supplementary Note 4).

Numerical simulations

To evaluate the performance of CNR-VD, we conduct various numerical simulations and compare its results among those obtained from noisy VD and other VD-based methods. To quantify the accuracy of estimators, we use the absolute error calculated as $\text{Error} = |\langle O \rangle^{\text{est}} - \langle O \rangle^{\text{ideal}}|$. For a fair comparison, we configure the same total measurement shot number for each method. We utilize second-order VD and perform the calibration procedure in each independent repeated experiment.

Our investigation begins with ascertaining the accuracy of CNR-VD. The stochastic Pauli gate noise model is utilized eliminating the need for RC technique. We define the noise level of the VD circuit as the magnitude of gate noise in the VD circuit, with the benchmark being the noise data from real machines⁴³ (i.e., when the noise level equals 1), and the value of the noise level indicates multiples of the benchmark noise value (See the detailed settings in the Supplementary Note 5).

We generate random noisy state $\rho_r = (1 - \epsilon)|\psi\rangle\langle\psi| + \epsilon|\psi_e\rangle\langle\psi_e|$ with the state preparation error rate ϵ by randomly choosing pure

states $|\psi\rangle\langle\psi|$ and $|\psi_e\rangle\langle\psi_e|$ which are mutually orthogonal. This formulation yields a straightforward fidelity measure^{19,44,45} of $\mathcal{F} = \text{Tr}[\rho_r|\psi\rangle\langle\psi|] = 1 - \epsilon$, providing a clear and direct assessment of the noise impact on the fidelity of the prepared state relative to the ideal state $|\psi\rangle\langle\psi|$.

We first carry out simulations of 4-qubit random states with a fixed state preparation error rate $\epsilon = 0.10$ and varying noise levels from 0.1 to 10.0, in comparison with ideal VD and noisy VD as illustrated in Fig. 3a. Results demonstrate that CNR-VD well recovers the outcomes obtained by ideal VD for low-weight observables or moderate noise levels. Our subsequent focus lies in the noise threshold at which the error-mitigated methods provide enhanced results compared to unmitigated estimation. With reference to the diagram in Fig. 3b, we indicate the corresponding noise thresholds with dashed lines. The increased noise threshold highlights the superiority of CNR-VD over noisy VD. In addition, we evaluate the failure rate of CNR-VD contingent on fluctuating noise levels and state preparation error rates. The failure rate signifies the probability that the results generated by CNR-VD are inferior to unmitigated estimation. Since the accuracy of unmitigated estimation is merely determined by the state preparation error, a larger state preparation error rate relative to the noise level of the VD circuit tends to result in a smaller failure rate, in accordance with the data shown in Fig. 3c. A boundary at failure rate 0.5 for CNR-VD is marked by a solid white line, signaling the point where CNR-VD's resilience to noise diminishes. The corresponding failure rate boundary for the noisy VD is also highlighted by a white dashed line. From our observations, CNR-VD extends the boundary to higher noise levels by up to one order of magnitude, which provides vital insights into the effectiveness and robustness of CNR-VD.

We also conduct a comparative study between CNR-VD and two common VD-based methods designed to counter noise in the VD circuit, which are Shadow Distillation²⁰ (SD) and noisy VD combined with zero-noise extrapolation (ZNE-VD)¹⁹. SD is a variation of VD that leverages shadow estimation to approximate the nonlinear function $\text{Tr}[\rho^2 O]/\text{Tr}[\rho^2]$ of ρ . ZNE-VD employs the zero-noise extrapolation (ZNE)^{7,12,46–51} QEM scheme to mitigate the noise in the VD circuit. The comparisons involve N -qubit random quantum states, which are generated by parameterized quantum circuits within the context of the Trotterized 1D Transverse-field Ising Model^{52,53} (See the detailed settings in Supplementary Note 5). In our simulations, we fixed the observable to be single Z Pauli observable σ_z^N on the last N -th qubit. For both CNR-VD and ZNE-VD, we employ the RC technique with 4 random instances in the VD circuit to enhance the effectiveness of the estimators. We assess the accuracy across a range of qubit numbers N from 2 to 8 and corresponding total measurement numbers M spanning from 10^3 to 10^5 . By specifically examining the accuracy at the extremes of these ranges, that is, using the maximum and minimum values of N and M , we aim to provide a clear illustration of the trend, highlighting how the algorithm's performance scales with the system size and measurement resources. Results are given in Fig. 4. We can observe that CNR-VD retains its superiority in accuracy even as the number of qubits increases. Notably, CNR-VD achieves an improvement in accuracy compared to noisy VD by more than an order of magnitude, as given in Fig. 4a. And in Fig. 4b, we can see that with the increasing number of shots, the performance of three VD-based methods (CNR-VD, ZNE-VD, SD) achieve

Fig. 3 | The accuracy of CNR-VD under Pauli gate noise with random states. **a** Results of 4-qubit random states under state preparation error rate $\epsilon = 0.10$ obtained from CNR-VD, ideal VD (VD with noiseless VD circuit) and noisy VD (VD with noisy VD circuit) with increasing noise levels of the VD circuit and weights of observables. **b** Accuracy of 4-qubit random states under state preparation error rate $\epsilon = 0.10$ shifting with an increase in the noise level. Yellow dots mark inflection points, dashed lines show noise thresholds, and gray dashed lines denote unmitigated estimation outcomes (Unmit). **c** CNR-VD's failure rate varies with noise levels and state preparation error rates. A white solid line at 0.5 failure rate marks CNR-VD's noise-resilience limit, with a white dashed line showing the boundary for noisy VD. All the data in the figure presents average results from 50 experiments with random states, with error bars for standard deviation.

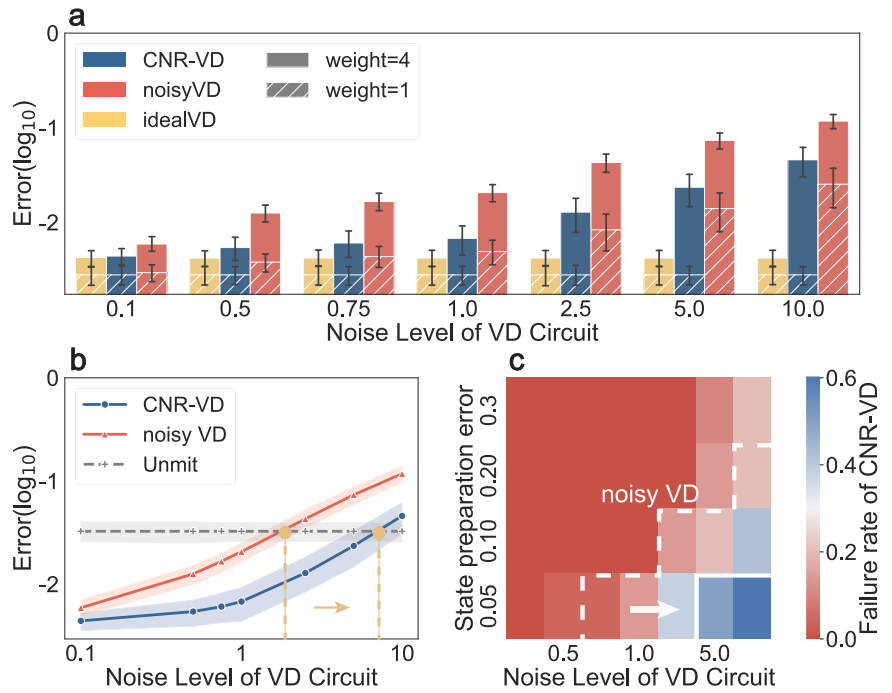
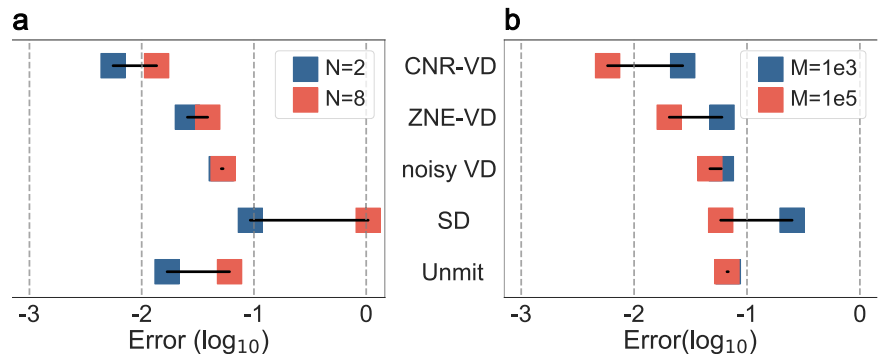


Fig. 4 | Comparison results of the expectation values of random parameterized states.

a Performance of five estimators: CNR-VD, ZNE-VD, noisy VD, SD, and unmitigated estimation (Unmit) with fixed measurement shots number $M = 2 \times 10^4$, and a number of qubits 2 and 8. **b** Performance of estimators for 4-qubit random parameterized state, with measurement shots number 10^3 and 10^5 . The displayed data represents the average outcome derived from 20 separate experiments for each qubit number and circuit depth. The chart includes error bars indicating the standard deviation.



significant enhancement in accuracy while the accuracy of noisy VD and unmitigated estimation are unable to further improve. It is noteworthy that among the three VD-based methods, CNR-VD consistently maintains a clear advantage.

Furthermore, Supplementary Note 5 presents simulation outcomes for higher-order VD using CNR-VD, demonstrating its capacity to recover the ideal behavior of third-order VD under moderate noise conditions. Considering practical scenarios involving circuit compilation, our investigation is extended within Supplementary Note 5 to assess the performance of VD-based algorithms, including the original VD, CNR-VD, and ZNE-VD, both pre- and post-compilation. The findings in Supplementary Note 5 indicate that, despite some variation post-compilation, CNR-VD maintains its superiority in performance among the VD-based algorithms.

Discussion

In conclusion, we propose the CNR-VD, which is established based on the original noisy VD and effectively overcomes the potentially harmful influence of gate noise in the VD circuit. The simulation results demonstrate that our proposed method can achieve better mitigation performance under diverse noisy conditions, in comparison to noisy VD and other VD-based methods designed against noise

in the VD circuit. Combined with the randomized compiling technique we can also increase the threshold for noise level at which QEM algorithms outperform unmitigated estimations.

Moreover, since our theoretical analysis is not limited to the VD algorithm, the CNR-VD estimator can also provide an effective solution to the practical challenge of circuit noise for the Hadamard-Test algorithms. This opens up vast research opportunities for our future work, such as exploring advanced techniques of CNR-VD in more complex error environments and investigating the practical implementation of our approach on real quantum computing platforms. We anticipate that this study may contribute valuable insights to the field of QEM and inform future research directions.

Methods

The implementation of the randomized compiling employed in the CNR-VD

We employ the randomized compilation (RC) technique in computing the CNR-VD estimator to come close to the main assumption of stochastic Pauli gate noise. In this section, we will provide a detailed explanation of the procedure.

Traditional RC is typically performed before and after the noisy multi-qubit gates and newly generated circuits exhibit logical equivalence with

their original counterparts. In practice, it is customary to select a fixed number of circuits with different random twirling instances and perform a certain number of measurements for each circuit, rather than resampling random twirling instances for per measurement shot.

We will use second-order VD for all subsequent discussions in this section. We can perform RC on the CSWAP chain and controlled- O chain separately. For the controlled- O chain, which is Clifford, we can randomly select a $2N + 1$ -qubit Pauli operator P and calculate the corresponding conjugated Pauli gate P^{twrl} by

$$P^{\text{twrl}} = CO \cdot P \cdot CO^\dagger, \quad (9)$$

where CO represents the controlled- O chain. When it comes to the CSWAP chain, as the CSWAP gate is not Clifford, the Pauli operator conjugated by the CSWAP gate may not still be a Pauli operator. We choose 8 Pauli gates from 3-qubit Pauli group in Table 1, which remain unchanged after being conjugated by CSWAP gate. Despite selecting only a small subset, RC yields promising results, as shown in Fig. 5. Since the CSWAP gates in the CSWAP chain only share the control qubit, we can readily derive the form of Pauli operators preserved by the CSWAP gate under conjugation

$$T = T^{\text{twrl}} = \{I, Z\}^0 \otimes \sum_{j=0}^{N-1} P^{j+1} P^{j+1+N}, \quad (10)$$

where P is the randomly chosen Pauli operator, and the superscript denotes the qubit on which the operator acts. It is possible to compile the Pauli twirling gates chosen for the CSWAP gate with the surrounding Pauli gates due to the property remaining unchanged under conjugation with the CSWAP gate. After compilation, only three additional layers of Pauli gates are needed for implementing RC in the VD circuit, as illustrated in Fig. 6. Additionally, the first layer of Pauli gates in the right figure can be compiled together with the single-qubit gates in the state preparation circuit of the input state. In the third layer of Pauli gates, it is only necessary to apply the Pauli gate on the first qubit, as we are only performing measurements on this qubit.

Table 1 | The chosen subset of 3-qubit Pauli operators, which are preserved by CSWAP gate under conjugation

q_0	q_1	q_2	q_0	q_1	q_2
I	I	I	Z	I	I
I	X	X	Z	X	X
I	Y	Y	Z	Y	Y
I	Z	Z	Z	Z	Z

Here, the q_i ($i = 0, 1, 2$) denotes the i -th qubit of the CSWAP gate, and I, X, Y, Z represent the identity, Pauli σ_x , Pauli σ_y , and Pauli σ_z operators, respectively.

Fig. 5 | The effect of randomized compiling (RC) with the chosen subset of 3-qubit Pauli operators, which are preserved by CSWAP gate under conjugation. **a** The original Pauli Transfer Matrix (PTM)⁵⁵ $\mathcal{P}$ for the composite noise channel Λ consists of depolarizing noise and amplitude damping noise. The elements in the heat-map represent the magnitude of PTM, which is calculated using $\mathcal{P}_{i,j} = \frac{1}{8} \text{Tr}[P_i \Lambda(P_j)]$ where P_i denotes the i -th 3-qubit Pauli operator. The off-diagonal elements of the PTM represent the non-Pauli components, which will be eliminated via RC. **b** The PTM for the composite noise channel after RC with the chosen subset of 3-qubit Pauli operators, which are preserved by CSWAP gate under conjugation. It can be observed that the off-diagonal elements have been effectively suppressed, although only a small subset is selected.

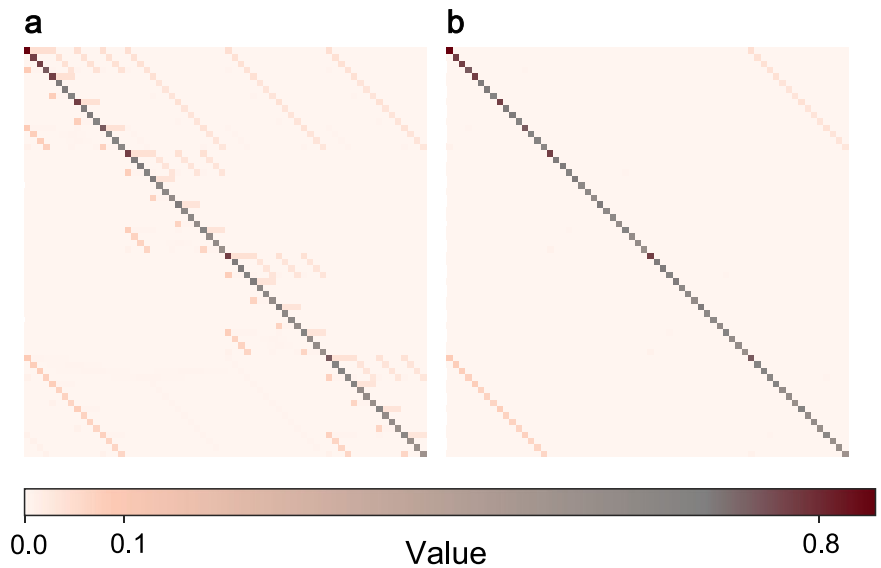
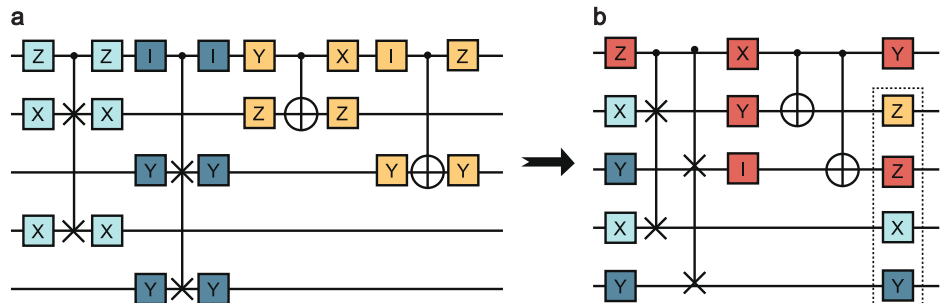


Fig. 6 | The compilation of Pauli twirling gates for second-order VD circuit. **a** The complete VD circuit with Pauli twirling gates. We have labeled the Pauli gates with different colors to indicate those associated with the same multi-qubit gate. **b** The VD circuit after compiling the Pauli twirling gates. The dashed rectangle in panel **b** signifies that the Pauli gates within it, corresponding to qubits other than the first, are not required to be executed, given that the VD algorithm only measures the first qubit.



Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Code availability

The code used for producing the results is available on the Zenodo repository⁵⁴.

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Author contributions

H.-L. H. supervised the whole project. X.-Y. X. and H.-L. H. conceived the idea and co-wrote the paper. X.-Y. X. provided the theoretical analysis and proofs. X.-Y. X., C. D., and H.-L. H. carried out the numerical simulations and analyzed the results. Z.S. and W.-S.B. contributed to the interpretation of results. All authors contributed to discussions of the results.

Competing interests

The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to Wan-Su Bao or He-Liang Huang.

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